

```
mysql -u root -p
```

```
CREATE DATABASE subquery_demo;
```

```
USE subquery_demo;
```

```
CREATE TABLE departments (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50)  
);
```

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    salary INT,  
    dept_id INT,  
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id)  
);
```

```
INSERT INTO departments VALUES  
(1, 'HR'),  
(2, 'IT'),  
(3, 'Finance');
```



```
INSERT INTO employees VALUES
```

```
(101, 'Aman', 30000, 1),
```

```
(102, 'Riya', 50000, 2),
```

```
(103, 'Karan', 45000, 2),
```

```
(104, 'Neha', 35000, 3),
```

```
(105, 'Raj', 60000, 2);
```

```
SELECT * FROM departments;
```

```
SELECT * FROM employees;
```

```
SELECT emp_name, salary
```

```
FROM employees
```

```
WHERE salary > (
```

```
    SELECT AVG(salary)
```

```
    FROM employees
```

```
);
```

```
SELECT emp_name
```

```
FROM employees
```

```
WHERE dept_id = (
```

```
    SELECT dept_id
```

```
    FROM departments
```

```
    WHERE dept_name = 'IT'
```

```
);
```

```
SELECT emp_name, salary
```

```
FROM employees
```

```
WHERE salary = (
```

```
    SELECT MAX(salary)
```

```
    FROM employees
```

);

```
SELECT emp_name
FROM employees
WHERE dept_id != (
    SELECT dept_id
    FROM departments
    WHERE dept_name = 'HR'
);
```

```
SELECT emp_name, salary
FROM employees
WHERE salary > (
    SELECT salary
    FROM employees
    WHERE emp_name = 'Neha'
);
```

```
SELECT dept_name
FROM departments
WHERE dept_id IN (
    SELECT dept_id
    FROM employees
    WHERE salary > 50000
);
```

```
SELECT emp_name
FROM employees
WHERE dept_id = (
    SELECT dept_id
    FROM employees
    WHERE emp_name = 'Raj'
);
```

```
SELECT MAX(salary) AS second_highest_salary
FROM employees
WHERE salary < (
    SELECT MAX(salary)
    FROM employees
);
```

```
EXIT;
```